Unreal Engine lecture final project

Goal

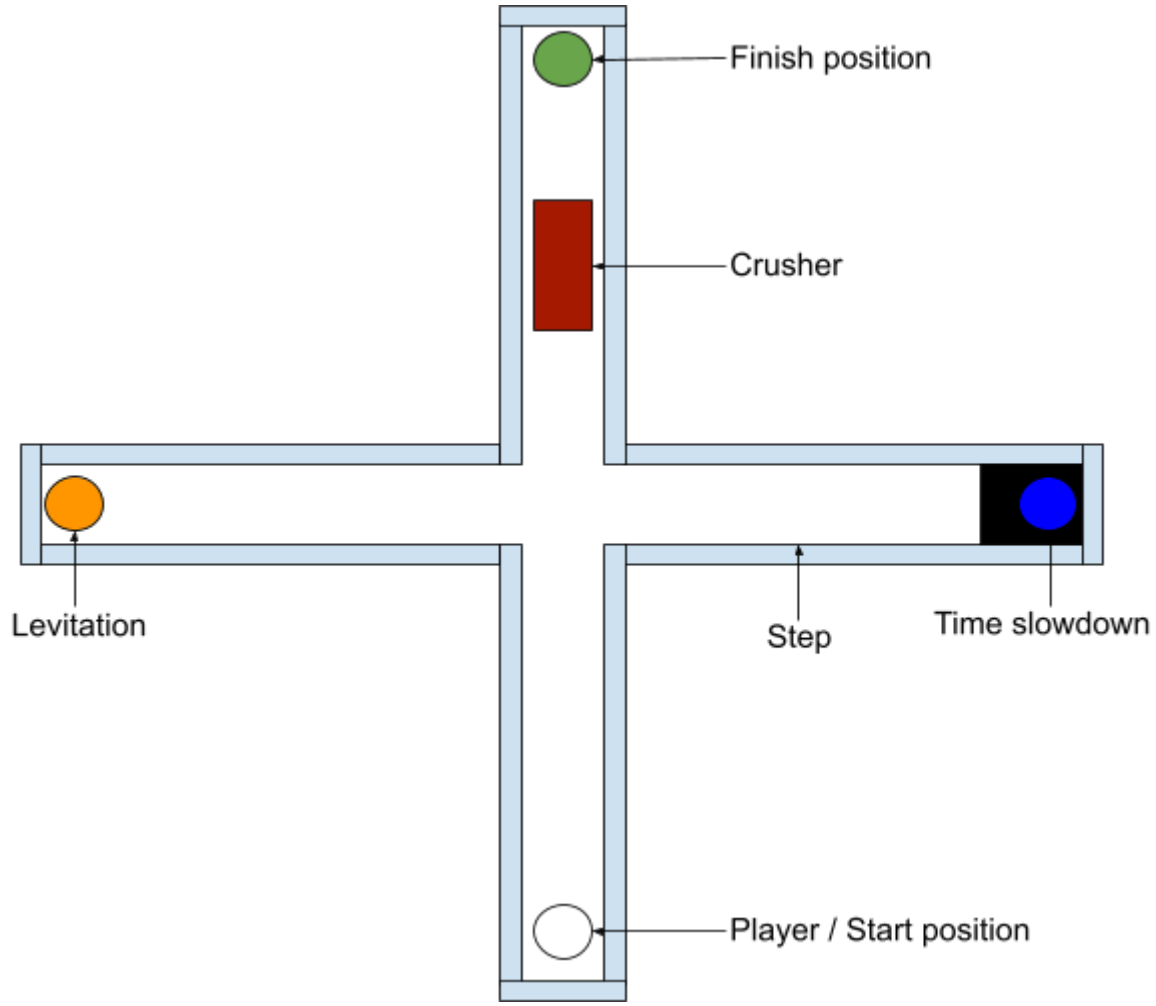
Create a simple game using Blueprints or C++ based on the following game design.

Game Design

In the game you navigate your player through a set of corridors, collecting abilities to overcome obstacles that prevent you from reaching the finish position.

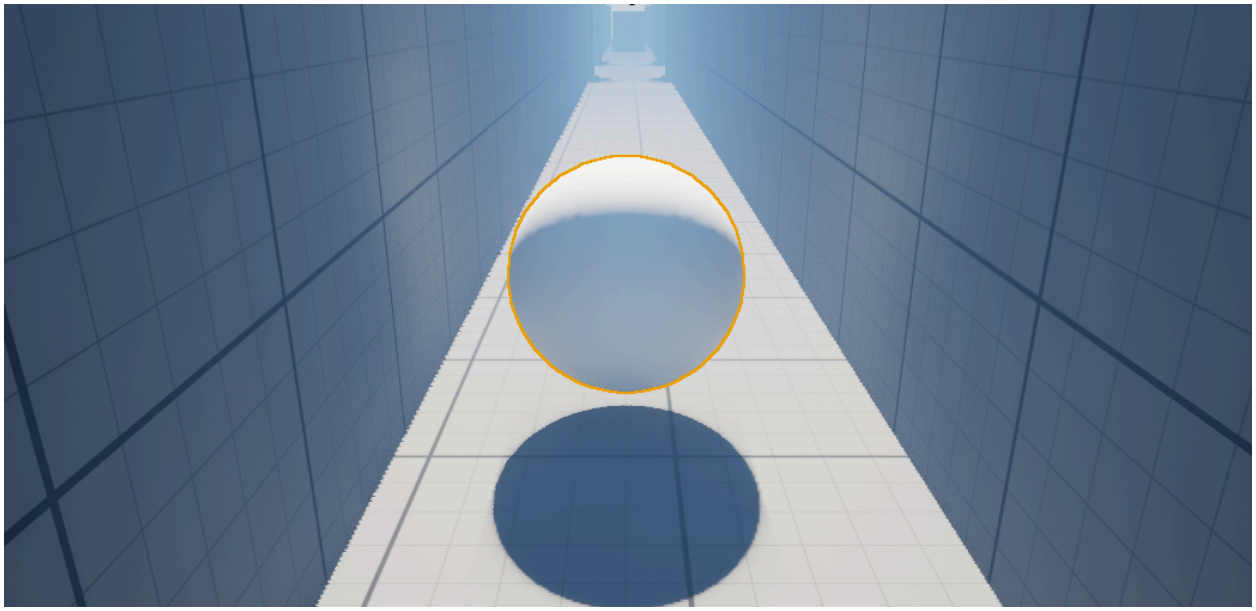
Map

The map is set of 4 corridors with walls high enough to prevent the Player from getting over them when using ability Levitation.



Player

The Player is a white sphere. The camera is set to third person, following the player rigidly. The Player starts at the start position marked on the map. The player collides with the walls of the corridors not being able to pass through them.



Controls

Key **W** - move the Player by certain distance (e.g. 1m) forward
Key **S** - move the Player by certain distance (e.g. 1m) backward
Key **A** - rotate the Player by 90 degrees counterclockwise
Key **D** - rotate the Player by 90 degrees clockwise
Key **E** - activate ability

Abilities

The Player can only own one ability at once. That means that upon achieving an ability, the Player loses any ability owned before.

Abilities are represented by spheres of different colors. The player starts to own the ability once the Player collides with the ability sphere. The ability sphere disappears upon collision with the Player.

Abilities active time is time limited. An ability can be activated only if it is not currently active.

Levitation

Orange sphere.

Allows the player to levitate for a limited time. Levitation means that the player is moved up by a certain height and stays there for the time of the ability. After the time of the ability is over, the player is moved to the original height before the levitation was triggered.

Levitation allows the player to fly over obstacle Step but does not allow the player to fly over obstacle Crusher.

Time slowdown

Blue color.

The sphere to collect the ability is placed on top of the obstacle Step.
Allows the player to slow down time without affecting the player for a limited time. Once activated, the speed of the obstacle Crusher slows down so that the player can pass under it.

Obstacles

Step

The Step is a simple static obstacle of black color that prevents the player from reaching the end of the corridor. The step is only so tall to prevent the player from passing it without the ability Levitation. Ability Levitation allows the player to fly over the Step.

Crusher

The Crusher is a dynamic obstacle of red color that moves up and down. When the player collides with the Crusher, the player is immediately moved to the Start position.

When the Crusher is at its high positions, there is enough space under it for the player to pass. The Crusher moves with such speed that the player normally cannot pass under it. Using ability Time slowdown slows down the Crusher so that it still moves but the player has enough time to pass under it when it is in its high position.

Game end

The Finish position is represented by a green sphere.
The game ends once the Player collides with the Finish position sphere.